Take-Home Exam, Advanced Microeconomics

Problem 1: Overworked Americans – Q1, Q2, Q4, Q5, Q7

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May 2026

Plan and storyline. Q1 – there is an inconsistency in the data. The US–EU hours-per-worker gap is real (panels (a)/(b)) but its cross-country relation to the tax wedge is loose: the 2023 OLS slope is signed correctly (-9.9 hrs/pp) but $R^2 = 0.16$ and the residual SD is $\approx$ 158 hours – one third of the entire DE–US gap (panel (c)). Taxes account for some of the cross-country variation, not most of it. **Q2 – the calibrated model fits, but only at an α above what micro evidence delivers; at the micro-consistent α, taxes explain only half the gap.** The Prescott log–log model gives $h^*(\tau) = (1-\tau)/(\alpha+1-\tau)$. At $\alpha^{\text{macro}} = 1.32$ (calibrated to $h^{US} = 0.33$) the model fits the cross-section. Restricting α to the micro-consistent $\alpha^{\text{micro}} = 0.28$ (such that $\varepsilon^M = 0.30$ at the US point), the Prescott-only prediction is a 13.4% US–EU log-drop vs. data 29.0% – so **taxes alone explain $\approx$46% of the cross-country hours gap**, leaving a residual 54% ($\sim$15.6 pp in logs) that the Prescott mechanism cannot account for; §4–§5 plus Q7 culture propose institutions and culture as candidate explanations and test whether they can absorb it. **Q4 – hours regulation:** under a low-η_0 RTM no-regulation baseline (the same machinery as §5 at small $\eta_0 \geq 0$, with $\eta_0 = 0$ nesting firm-power), a binding cap is welfare-improving iff $\bar{h} \in (2h^{\text{soc}} - h(\eta_0),\, h(\eta_0))$. The band is non-empty only for $\eta_0 < \eta^{\text{soc}} \approx 0.41$: *the cap and the union are substitutes.* Manning (2003) monopsony and Cahuc–Carcillo (2014) sticky contracts both map to low effective η_0. **Q5 – single union: canonical right-to-manage (Nickell 1982; LNJ 1991).** Union and firm Nash-bargain over wage w; firm sets $h = h^d(w)$ on labour demand. As η rises, w rises and the firm cuts h along its labour-demand curve. **The welfare verdict depends on the welfare criterion (the weight given to worker vs. firm surplus).** Under realistic Saez–Stantcheva (2016) inequality aversion ($\lambda = 0.30$; HSV 2017 + log utility), $\eta^* \approx 0.58$ – *a moderately strong union is welfare-optimal* (§5.5b). The equal-weight knife-edge $\lambda = 0$ gives the corner $\eta^* = 0$, but this assigns no value to the firm-to-worker redistribution and is not a defensible default. Applied to a two-sector economy with heterogeneous productivity, the *same* RTM machinery delivers the canonical single-sector story in the good sector (positive shock $\rightarrow$ union extracts wage rents $\rightarrow$ hours rise less than under free market) and an extra asymmetry in the bad sector (negative shock $\rightarrow$ wages cannot fall under Bewley 1999 / Holden 1994 downward rigidity $\rightarrow$ hours and profit absorb the entire shock). The good-sector outcome is the single-sector RTM result; the bad-sector outcome is what the same model produces once you allow the wage to be downward-sticky. **Good shocks pass through diluted; bad shocks pass through amplified. Q7 – welfare comparison: institutional channels.** The labour-market verdict combines the Q4 cap and Q5 union channels with a Barro-style social-productivity correction $A^{\text{soc}}(h) = A_0[1 + \eta_\phi(h/h^{US} - 1)]$ that credits hours with their public-goods, capital, on-the-job-HC and R&D externalities (refs Aschauer 1989, Trabandt–Uhlig 2011, Bils–Klenow 2000, Rachel 2020); central calibration $\eta_\phi = 0.40$. Verdict: under equal weights ($\lambda = 0$) US welfare-*dominates* because EU institutions pull h below the social optimum and the social-productivity drag compounds. Under realistic Saez–Stantcheva (2016) $\lambda = 0.30$, EU welfare-*dominant* by $\Delta W^\lambda \approx +0.10$ – union-driven redistribution overwhelms the misallocation cost. Dominant US–EU welfare differences economy-wide come from non-labour channels (Jones–Klenow 2016). Reproducible: `research/build_figure_panel.py` (Q1 figure); `research/sim_models_v2.py` (Q2 decomposition, Q5 RTM and §5.5b Saez–Stantcheva, §5.6 sectoral shocks, Q7 cultural expansion, layered framework, §7.5 sensitivity).

1. Q1 – The hours gap and its noisy relation to tax wedges

Question 1. *Document the difference between hours worked in the US vs Europe in recent years.*

1.1 Definitions

For the working-age (15–64) population: hours per employed worker HE (OECD `DSD_HW`, the intensive object the labour-supply model speaks to), employment rate $e \equiv E/N$, hours per adult $H/N \equiv HE \cdot e$. The Bick–Brüggemann–Fuchs-Schündeln (2019) decomposition $\Delta \log(H/N) = \Delta \log HE + \Delta \log e$ assigns roughly half of the EU–US gap to each margin; we focus on HE.[1]

1.2 The fact

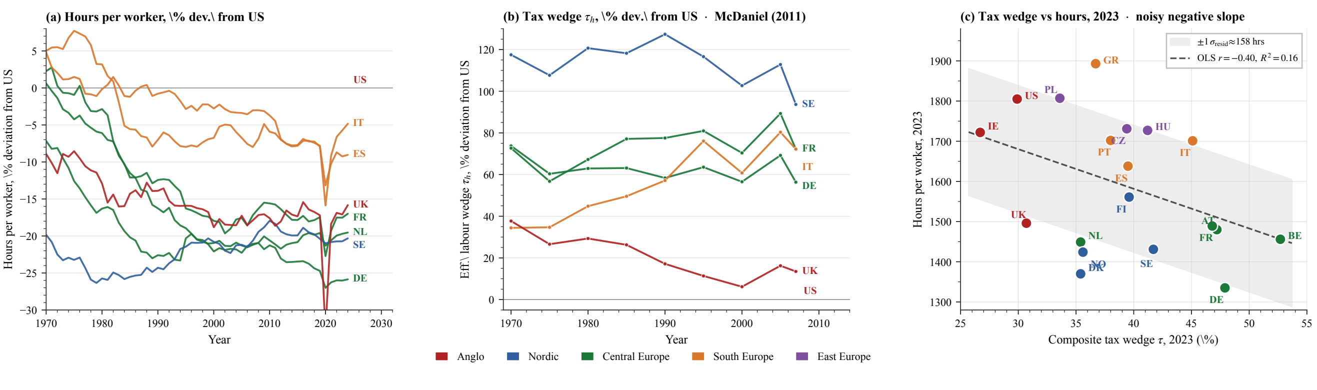

Figure 1: **Three-panel evidence.** (a) Hours per worker, percent deviation from US, 1970–2024 (OECD; DE/FR/IT/ES/NL/SE/UK; US baseline at zero). (b) Effective labour wedge τ_h, percent deviation from US, 1970–2007 (McDaniel 2011 national-accounts measure). (c) Labour-tax wedge τ (OECD) vs hours per worker, 2023 cross-section ($n = 19$ OECD countries); OLS line and $\pm 1\sigma_{\text{resid}}$ band. *Source / replication:* `research/build_figure_panel.py`.

Three readings: **(i)** Hours fell in every focus country relative to the US since 1970 (DE/NL ≈ -25 to -30%; IT/ES ≈ -10 to -15%). **(ii)** The labour wedge rose in every focus country (SE +120%, DE/FR/IT +60–70% above US in 2007) – two

[1] Cross-country e differences are substantial (US 71% vs IT 59% in 2019); the prompt asks about hours, so HE is the body's object.

series moving in opposite directions, qualitatively consistent with a tax-driven story. **(iii)** The 2023 cross-section is signed correctly but very noisy: OLS slope -9.9 hrs/pp, $R^2 = 0.16$, residual SD ≈ 158 hours. To scale: the entire DE–US gap is 470 hours, so the residual SD is one third of what we want to explain. Italy ($\tau = 45.1\%$) and Greece (36.7%) work *more* hours than France (46.8%); the four highest-τ countries (BE, DE, AT, FR) span ≈ 200 hours within a 6pp wedge band.

1.3 The inconsistency the rest of the problem is asked to explain

Taxation alone fits poorly: $R^2 = 0.16$ and a residual SD of one third of the gap to explain. The remainder of the paper searches for the missing channels (§2–§7). **Time-use caveat.** Aguiar–Hurst (2007, *QJE* 122(3)) show that conventional "hours worked" miss home production, which behaves differently across countries. Rogerson (2008, *JPE* 116(2)) exploits this margin; we treat it as outside the scope of this submission but flag it as the canonical first-order extension.

Identification. The OLS slope is descriptive, not causal: omitted institutions (union coverage, hours regulation), reverse causality (preferences for leisure shift both h and τ; Q7), and country fixed effects bias $\hat{\beta}$. The structural model of §2–§7 treats the cross-section as a moment to match, not as a causal slope.

2. Q2 – Prescott model and the elasticity that would be needed

Question 2. *Summarise the main differences in income tax between US and Europe. Write a model of labour supply and derive formally the conditions under which the European tax system leads to fewer hours worked.* Use the words 'hence' and 'novel' twice.

2.1 Modelling choice

Log–log preferences with a balanced lump-sum rebate (Prescott 2004; Cooley–Prescott 1995): one-parameter (α) closed-form $h^*(\tau)$ and a clean elasticity comparative static. Static (no Frisch margin), balanced budget (income effect rebated lump-sum), abstracts from capital (small for prime-age workers, CGM 2011) and heterogeneity (Q4 treats heterogeneous workers).

2.2 The simplest one-margin model

Log–log preferences $u(c, h) = \log c + \alpha \log(1-h)$, wage w, and a proportional labour-tax wedge τ. The FOC at the symmetric equilibrium $c = wh$ gives

$$\boxed{h^*(\tau) \;=\; \frac{1-\tau}{\alpha+1-\tau}, \qquad \varepsilon^M_{h,1-\tau} \;=\; \frac{\alpha}{\alpha+1-\tau} \in (0,1).} \tag{1}$$

h^* is monotone in τ – the cross-section sign of panel (c) becomes a structural prediction with one free parameter α.

2.3 Parametric version, anchor, and what it would take to fit the cross-section

Anchor. Anchoring $h^{US} = 0.33$ at $\tau^{US} = 0.35$ (the BLS-OECD ratio of weekly hours-worked to total awake hours) inverts (1) to give $\alpha = (1-\tau)(1-h^*)/h^* = 1.32$ (Prescott 2004 uses a similar calibration). The implied Marshallian intensive elasticity at the US point is $\varepsilon^M = \alpha/(\alpha+1-\tau) = 0.67$.

The model calibrated at $\alpha = 1.32$ delivers $\varepsilon^M \in [0.67, 0.75]$ along the supply curve (rising in τ). Predicted EU hours at $\tau = 0.56$ are $h^{EU} = 0.250$ against data 0.247: *the calibrated model essentially fits the cross-section.* The puzzle is that the implied $\varepsilon^M \in [0.67, 0.75]$ sits $\sim 2.5\times$ above the microeconometric Hicksian intensive of ≈ 0.30.

Elasticity taxonomy. CGM (2011) report Marshallian intensive ≈ 0.30 for prime-age men and Hicksian intensive ≈ 0.30 – the micro consensus. Hansen (1985)/Rogerson (1988) indivisible-labour Frisch $\approx$ 1–3 is the macro requirement. The calibrated $\varepsilon^M \in [0.67, 0.75]$ sits between, $\sim 2.5\times$ above the micro Hicksian. Three modern developments shrink but do not eliminate the gap: life-cycle aggregation (Rogerson–Wallenius 2009; Erosa–Fuster–Kambourov 2016) lifts the aggregate elasticity to 0.7–1.0; tax progressivity (Heathcote–Storesletten–Violante 2017) is a separate cross-country margin; participation extensive elasticity (BFL 2018) compounds with the intensive. *Hence* the cross-section fits but at an α inconsistent with quasi-experimental evidence (Saez 2001; Chetty 2012) – the substantive content of the Prescott–CGM controversy. *Sufficient-statistics bridge*: Saez (2001) gives the optimal top rate $\tau^* = 1/(1 + a\,e^H)$; with CGM's micro $e^H \approx 0.30$ and Pareto $a \approx 1.5$ this is the canonical Diamond–Saez (2011) headline $\tau^* \approx 69\%$, broadly in line with European top rates. Reconciling Prescott's larger uncompensated elasticity would push τ^* down – a tension HSV (2017) resolve by adding progressivity as a separate margin.

2.4 Decomposition at the micro elasticity

At $\alpha = 1.32$, α absorbs everything (preferences, institutions, culture). Restricting α to a micro-consistent value isolates the share of the gap that the Prescott mechanism can no longer carry. Solving $\varepsilon^M = \alpha/(\alpha+1-\tau) = 0.30$ at $\tau^{US} = 0.35$ gives $\alpha^{\text{micro}} = 0.279$. Predicted hours: $h^*_{US} = 0.700$, $h^*_{EU} = 0.612$, log-drop -13.4%. Data log-drop is -29%. **Taxes alone produce $\approx 46\%$ of the observed gap; the residual 54% (15.6pp in logs) lies outside the Prescott mechanism, and its source is not yet identified.** §4–§5 plus Q7 culture propose three candidate channels (hours regulation, unions, culture) and test whether each can absorb part of it.

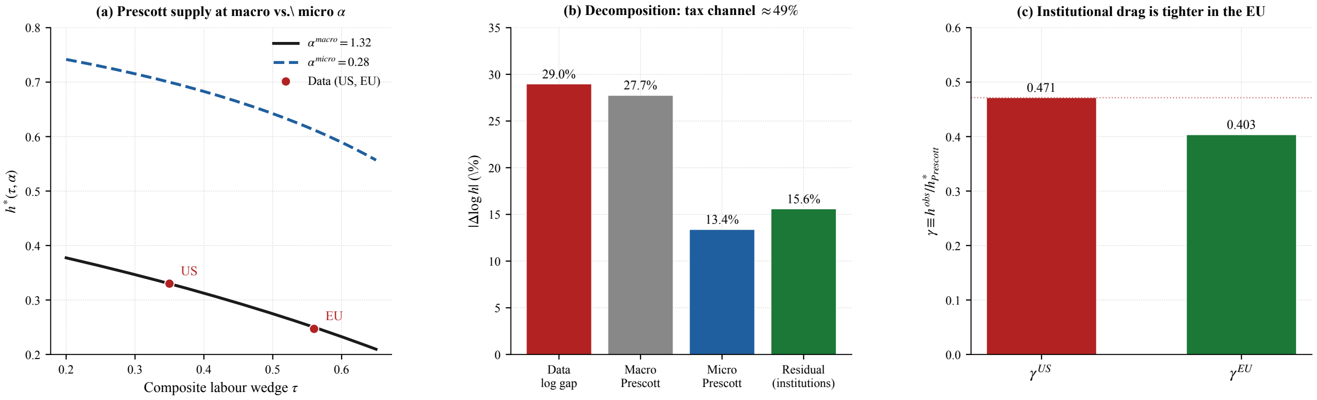

Figure 2: **Q2 macro–micro decomposition.** **(a)** Prescott supply curves at $\alpha^{\text{macro}}=1.32$ and $\alpha^{\text{micro}}=0.279$; US/EU data marked. **(b)** Decomposition of the 29% data gap: Prescott-at-micro explains 13.4% (46% share), unexplained residual 15.6% (54% – to be addressed in §4–§5 plus Q7 culture). **(c)** Unexplained-shortfall ratio $\gamma \equiv h^{obs}/h^*_{\text{Prescott}}(\alpha^{\text{micro}})$ – the share of Prescott-predicted hours that are actually worked. $\gamma^{US}=0.471$, $\gamma^{EU}=0.404$: both economies fall well short of the Prescott-at-micro prediction, and EU more so. The gap between γ and 1 is what §4–§5 plus Q7 culture have to absorb. *Source:* `sim_models_v2.py`.

Reading panel (c). Two things stand out. First, the *absolute* level of γ is implausibly low for both countries – workers do roughly 40–47% of the hours Prescott-at-micro predicts. This is *not* a Europe-vs-US phenomenon: it tells us that at the micro elasticity, the Prescott model in absolute levels is too steep to be taken seriously as a level prediction (it implies $h^{US} \approx 0.70$, which is workers spending 70% of their awake time at work). The useful content of the Prescott calibration is therefore the *cross-country log ratio*, not the level – which is exactly what the 46/54 decomposition uses. Second, the *ratio* $\gamma^{EU}/\gamma^{US}=0.404/0.471=0.858$ tells us how much more shortfall the EU has *relative to* the US: about 14% of Prescott-predicted hours are missing in EU beyond what is already missing in US. **This ∼14% extra cross-country shortfall is the precise quantity that an institutional or cultural candidate channel must produce** – not the absolute 50–60% level shortfall, which is a feature of the Prescott model itself. §4–§5 each test whether their mechanism can generate a US–EU h-gap of this size.[2]

3. Common framework for Q4–Q5–Q7

3.1 Primitives

We adopt a quadratic-disutility worker, linear consumption, and a linear-quadratic firm-profit function – the simplest closed-form specification that admits both worker- and firm-preferred hours and a sign-determinate welfare condition:

$$V_w(h) = (1-\tau)wh - \tfrac{\alpha}{2}h^2,\quad h^* = \frac{(1-\tau)w}{\alpha}; \qquad V_f(h) = Ah - \tfrac{\beta}{2}h^2 - wh,\quad h^{\text{firm}} = \frac{A-w}{\beta}.$$

Joint surplus $W(h) \equiv V_w(h) + V_f(h)$ is concave with maximum

$$h^{\text{soc}}(\tau,\alpha,A,\beta,w) \;=\; \frac{(1-\tau)w + (A-w)}{\alpha+\beta}.$$

The setup is shared by §4 (regulation), §5 (unions), and Q7 (culture). w is partial-equilibrium; heterogeneous workers are addressed at the end of §4.

3.2 Baseline calibration

$A = 2$, $w = 1$, $\alpha = \beta = 2$, $\tau = 0.4$ gives $h^* = 0.30$, $h^{\text{soc}} = 0.40$, $h^{\text{firm}} = 0.50$. Only the ordering $h^* < h^{\text{soc}} < h^{\text{firm}}$ is load-bearing; qualitative results invariant to α,β separately (`sim_mechanisms.py`). At common τ and preferences neither Walrasian clearing nor low-η_0 RTM generates a US–EU gap; three channels break the symmetry: hours regulation (§4), unions (§5), and culture (Q7).

4. Q4 – Hours regulation: the optimality condition for a cap

Question 4. *Labour-market regulation limiting hours. Welfare effects on workers and firms?*

4.1 Modelling choice

A cap is a quantity instrument: welfare-improving if it pulls hours toward h^{soc}, harmful if it overshoots below. The closest sufficient-statistics analogue is Lee–Saez (2012) on the minimum wage, but their welfare logic runs through extensive-margin rationing whereas a binding hours cap rations no one on the extensive margin (every worker still works); it rations low-α workers on the intensive margin (§4.6). Micro-foundation: sectoral collective contracts that fix weekly hours and are not renegotiated as productivity shifts (Cahuc–Carcillo 2014). §4.2 derives the no-regulation equilibrium from a low-η_0 Nash bargain – sticky contracts and Manning (2003) monopsony both map to low effective η_0.

[2] Sectoral contract stickiness (Cahuc–Carcillo 2014) – weekly hours fixed by collective contract not renegotiated as τ changes – is captured implicitly by the cap mechanism of §4 in the static framework.

4.2 Setup: low-η_0 RTM *derives* the distortion

The Q4 cap analysis needs a no-regulation reference at which equilibrium hours sit *above* h^{soc} – otherwise there is nothing for the cap to correct. Rather than *stipulating* "firm picks h^{firm} and worker accepts," we run the same Nash bargain as §5 at low worker weight $\eta_0 \geq 0$: union and firm bargain over w, the firm then sets $h = h^d(w) = (A-w)/\beta$ on its labour demand. The $\eta_0 \to 0$ corner nests the firm-power axiom ($w=1$, $h=h^{\text{firm}}=0.5$, worker IR-binding); for any $\eta_0 > 0$ the equilibrium $(w(\eta_0), h(\eta_0))$ *emerges from the bargain*, removing the load-bearing "firm picks" axiom. Manning (2003, *Monopsony in Motion*, Princeton; review in Manning 2021, *ILR Review* 74(1); empirical evidence in Card–Cardoso–Heining–Kline 2018, *JOLE* 36(S1)) and Cahuc–Carcillo (2014) sticky collective contracts both map to low effective η_0 in this framework: each is a complementary microfoundation, not a competing one. With cap $\bar{h}$ the constrained equilibrium is

$$h^{eq}(\bar{h};\eta_0) \;=\; \min\{\bar{h},\; h(\eta_0)\}.$$

4.3 Optimality condition (Q4 result)

Claim. Total surplus $W(h)$ is strictly concave and peaks at h^{soc}. Comparing the constrained equilibrium to the no-cap baseline at bargaining weight η_0:

$$\boxed{\Delta W(\bar{h};\eta_0) \;\equiv\; W(\bar{h}) - W(h(\eta_0)) \;\begin{cases} =0 & \text{if } \bar{h} \geq h(\eta_0) \\ >0 & \text{iff } \bar{h} \in \big(2h^{\text{soc}} - h(\eta_0),\, h(\eta_0)\big) \\ =0 & \text{at } \bar{h} = 2h^{\text{soc}} - h(\eta_0) \\ <0 & \text{if } \bar{h} < 2h^{\text{soc}} - h(\eta_0) \end{cases}} \tag{2}$$

provided $h(\eta_0) > h^{\text{soc}}$; if $h(\eta_0) \leq h^{\text{soc}}$ the welfare-improving region is empty and any binding cap reduces W. The cap attains its maximum at $\bar{h} = h^{\text{soc}}$ (whenever the band is non-empty), with peak gain $\frac{1}{2}(\alpha+\beta)\big(h(\eta_0) - h^{\text{soc}}\big)^2$ above the no-cap level.

Threshold η^{soc} for the cap to be useful at all. Substituting $h(\eta) = h^{\text{soc}} = 0.40$ (i.e. $x = 0.8$) into the Q5 quadratic $1.1x^2 - 0.6(2-\eta)x + 0.1(1-\eta) = 0$ gives $0.704 - 0.48(2-\eta) + 0.1(1-\eta) = -0.156 + 0.38\eta = 0$, so

$$\eta^{\text{soc}} \;\approx\; 0.41.$$

For $\eta_0 < \eta^{\text{soc}}$ the welfare-improving band is non-empty and a well-chosen cap raises welfare; for $\eta_0 \geq \eta^{\text{soc}}$ the bargain has *already* pulled h to or below h^{soc} and any binding cap destroys welfare.

At $\alpha = \beta$ and $\eta_0 = 0$, the band reduces to $\bar{h} \in (h^*, h^{\text{firm}}) = (0.30, 0.50)$ (textbook symmetric expression). **Proof.** $W(h)$ is quadratic with second derivative $-(\alpha+\beta)$; $\Delta W(\bar{h};\eta_0)$ is concave with roots at $h(\eta_0)$ and the mirror point $2h^{\text{soc}} - h(\eta_0)$. □

4.4 Numerical simulation

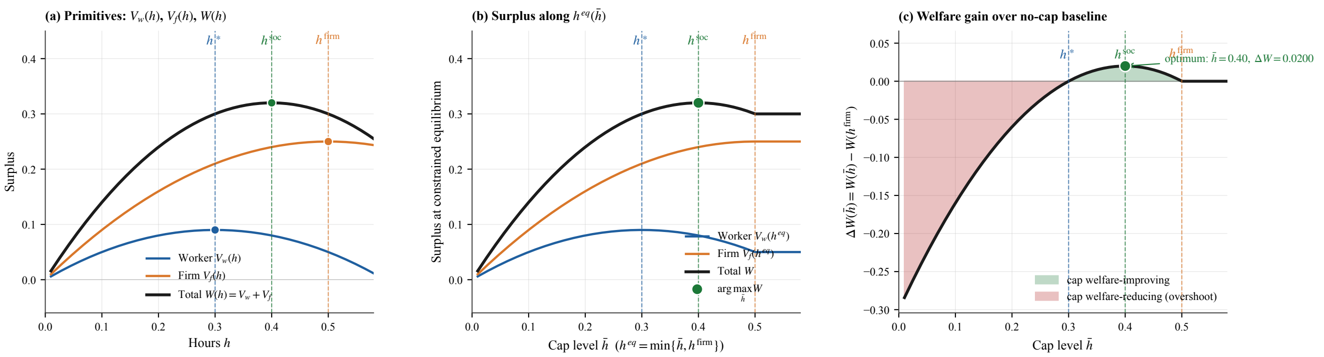

Figure 3: **Hours-cap simulation.** **(a)** primitives V_w, V_f, W over h with interior maxima marked; **(b)** surplus along $h^{eq}(\bar{h}) = \min\{\bar{h}, h^{\text{firm}}\}$; **(c)** $\Delta W(\bar{h})$ vs no-cap baseline – green band is the welfare-improving region of (2), red is overshoot. Maximum gain $\Delta W = 0.020$ at $\bar{h} = h^{\text{soc}} = 0.40$. *Source:* `sim_mechanisms.py`.

4.5 Empirical anchor and US–EU mapping

The 35-hour Aubry reform (France 1998–2000) is the canonical anchor: Estevão–Sá (2008) and Chemin–Wasmer (2009) report $\approx$4% aggregate hours decline in covered sectors with near-zero employment effect. Mapping: $\bar{h}^{Aubry} \approx 0.875 \cdot h^{\text{firm}} = 0.4375$, above $h^{\text{soc}} = 0.40$ – inside the welfare-improving band. US has no binding cap (FLSA 40-hour rarely binds at the median).

Cap and union are substitutes, not complements. The band (2) shrinks as η_0 rises and vanishes at $\eta^{\text{soc}} \approx 0.41$. In an economy where unions have already pulled h to h^{soc}, a binding cap on top destroys welfare. This rationalises why the 35-hour cap was contentious in France (RTM coverage $\sim$95%): at high η_0 the cap fights a union channel already doing the work. The US ($\eta \approx 0$) would benefit from a binding cap; EU North (DE/SE) at high η would not.

4.6 Heterogeneous workers: who is rationed

With worker types indexed by α_i, every worker is dictated $h^{\text{firm}} = 0.5$ under firm-power and forced to $\bar{h}$ by a binding cap. The payoff change $\Delta V_w^i(\bar{h}) = (\bar{h} - h^{\text{firm}})[(1-\tau)w - \frac{\alpha_i}{2}(\bar{h} + h^{\text{firm}})]$ is positive iff $\alpha_i > \alpha^*(\bar{h})$ where

$$\boxed{\alpha^*(\bar{h}) \;=\; \frac{2(1-\tau)w}{\bar{h} + h^{\text{firm}}}.} \tag{3}$$

At $\bar{h} = h^{\text{soc}} = 0.4$: $\alpha^* = 1.333$. High-α (leisure-loving) workers gain; low-α (income-loving) workers are rationed on the intensive margin. **Aggregate sign condition.** Integrating across the population, the cap is welfare-improving on average iff

$$\int \Delta V_w^i(\bar{h})\, dF(\alpha) \;+\; \Delta V_f(\bar{h}) \;>\; 0 \quad\Longleftrightarrow\quad \bar{\alpha} \;>\; \alpha^*(\bar{h}) \;-\; \frac{2\Delta V_f(\bar{h})}{(h^{\text{firm}} - \bar{h})^2},$$

i.e. a population whose mean leisure-preference $\bar{\alpha}$ exceeds the threshold $\alpha^*(\bar{h})$ adjusted by a firm-side correction. At baseline $\bar{\alpha} = 2$ and $\alpha^* = 1.333$ at $\bar{h} = h^{\text{soc}}$, so the cap is welfare-improving on average; a country with median $\alpha = 1$ (income-loving workforce) would lose.

5. Q5 – Single union in a two-sector economy

Question 5. *Unions in the economy and their response to sectoral shifts. Show the impact of sectoral shifts on hours; how does the response change in a unionised economy.*

5.1 The two-sector economy

The economy has two sectors $j \in \{\text{G}, \text{B}\}$ with productivities $A_{\text{G}} > A_{\text{B}}$ (good and bad state) and population shares s, $1-s$. Within each sector a representative worker–firm pair has the quadratic-linear primitives of §3:

$$V_w(w_j, h_j) = (1-\tau)w_j h_j - \tfrac{\alpha}{2} h_j^2, \qquad V_f(w_j, h_j; A_j) = A_j h_j - \tfrac{\beta}{2} h_j^2 - w_j h_j.$$

A single union covers both sectors; in each, the firm sets hours on its labour-demand curve $h_j^d(w_j) = (A_j - w_j)/\beta$ ex post. Three wage-setting regimes are admissible:

(a) **free market**: no union; firm-power Nash bargain ($\eta = 0$) at each A_j with worker IR-binding;
(b) **flexible union**: sector-specific RTM bargain with worker weight $\eta > 0$;
(c) **rigid union**: downward wage rigidity (Bewley 1999; Holden 1994) – wages can be revised *up* but not below the pre-shock benchmark.

Baseline: $\alpha = \beta = 2$, $\tau = 0.4$, $V_w^0 = 0.05$, $A_{\text{G}} = 2.2$, $A_{\text{B}} = 1.8$ (a symmetric $\delta = 0.2$ shock around a pre-shock symmetric productivity $A_{\text{pre}} = 2$). The single-state result at $A = 2$ is the symmetric pre-shock benchmark.

5.2 The union bargain in each sector

We adopt the canonical right-to-manage Nash bargain (Nickell 1982; Layard–Nickell–Jackman 1991): in each sector j, union and firm bargain over w_j and the firm sets h_j on labour demand. Per-sector Nash bargain:

$$w_j(\eta; A_j) = \arg\max_{w} \,[V_w(w, h_j^d(w)) - V_w^0]^{\eta}\, V_f(w, h_j^d(w); A_j)^{1-\eta}.$$

Let $x_j \equiv A_j - w_j$, so $h_j = x_j/\beta$. The Nash FOC reduces to a sector-specific quadratic:

$$\boxed{a\, x_j^2 - b_j(2-\eta)\, x_j + c(1-\eta) = 0, \quad a = \tfrac{2(1-\tau)}{\beta} + \tfrac{\alpha}{\beta^2}, \;\; b_j = \tfrac{(1-\tau)A_j}{\beta}, \;\; c = 2V_w^0,} \tag{4}$$

with $(a, c) = (1.1, 0.1)$ at baseline; $b_{\text{G}} = 0.66$, $b_{\text{B}} = 0.54$. The mechanism is the same in both sectors: as η rises, w_j rises, the firm's marginal cost of labour rises, and the firm slides down its labour-demand curve to a lower h_j. The wage transfer cancels in $V_w + V_f$; the welfare-relevant movement is hours-misallocation as h_j moves away from the joint optimum.

5.3 Welfare criterion: equal weights vs Saez–Stantcheva λ

The equal-weight utilitarian $W = V_w + V_f$ values *nothing* of the firm-to-worker redistribution that the union effects – the wage transfer cancels between the two parties. The Saez–Stantcheva (2016) generalised welfare weights formalise the alternative, applied per sector and aggregated linearly:

$$\boxed{W_j^{\lambda}(\eta) = (1+\lambda)V_w^j(\eta) + (1-\lambda)V_f^j(\eta), \qquad \bar{W}^{\lambda}(\eta; s) = s\, W_{\text{G}}^{\lambda} + (1-s)\, W_{\text{B}}^{\lambda},}$$

with $\lambda \in [0, 1]$ the redistribution premium ($\lambda = 0$ equal weights, $\lambda = 1$ Rawlsian). Three independent anchors converge on $\lambda \in [0.25, 0.45]$: HSV (2017) $\tau_p = 0.18$ under log utility $\Rightarrow \lambda \approx 0.43$; OECD worker/capital-owner consumption gap $\sim$ 4:1 $\Rightarrow \lambda \approx 0.30$; Diamond–Saez (2011) population-average in $[0.3, 0.5]$. Central case: $\lambda = 0.30$.

Differentiating $W_j^{\lambda}(\eta)$ along the bargained path yields the closed-form welfare-maximising x_j^* per sector:

$$\boxed{x_j^*(\lambda) = \frac{0.3\, A_j(1+\lambda)}{0.6 + 1.6\lambda}, \quad h_j^* = x_j^*/\beta, \quad w_j^* = A_j - x_j^*,} \tag{5}$$

with $\eta_j^*(\lambda)$ recovered from (4). At baseline and $\lambda = 0.30$: $\eta^* \approx 0.58$ in each sector (mild A_j-dependence). The aggregate welfare-maximising η at $s = 0.5$ is essentially the same.

5.4 Asymmetric pass-through: the same bargain produces two effects

The same Nash bargain delivers *different equilibria across the two sectors*, and the union's downward wage rigidity adds a further asymmetry on the bad-sector side.

Good sector: union extracts wage rents. At $\eta = 0.5$ the bargain gives $w_{\text{G}} = 1.354$, $h_{\text{G}} = 0.423$ (vs. free-market $w_{\text{G}}^{\text{FM}} = 1.081$, $h_{\text{G}}^{\text{FM}} = 0.559$). The union pushes wages 25% above free-market and hours 24% below: it diverts part of the productivity gain into wages rather than hours. Worker rents rise ($V_w : 0.05 \to 0.165$), firm profit falls ($V_f : 0.313 \to 0.179$); the joint pie size is roughly preserved, only its split changes.

Bad sector under flexible bargain: shock partly absorbed by wages. At $\eta = 0.5$: $w_{\text{B}} = 1.132$, $h_{\text{B}} = 0.334$. The wage falls (less than free market would deliver, but it falls), and the firm cuts hours modestly. The shock is split between wages and hours.

Bad sector under rigid union: the wage cannot fall. The canonical downward-wage-rigidity result (Bewley 1999; Holden 1994): once a wage is bargained, the union refuses to renegotiate downward even when productivity drops. So the bad-sector wage is stuck at the pre-shock level $w_{\text{pre}} = 1.242$, hours collapse to $(A_{\text{B}} - w_{\text{pre}})/\beta = 0.279$, firm profit crashes to $V_f = 0.078$. *Hours fall* 36% *further than free market; firm profit falls* 60% *further.* The shock is concentrated entirely on the firm-and-hours margin.

Headline asymmetry. Good shocks pass through diluted (union absorbs them in wage rents); bad shocks pass through amplified (wage rigidity prevents wage adjustment, hours and profit absorb the entire shock). Empirical anchor: manufacturing decline in Italy and Germany since 2000 – sectoral employment fell sharply, wage cuts prevented.

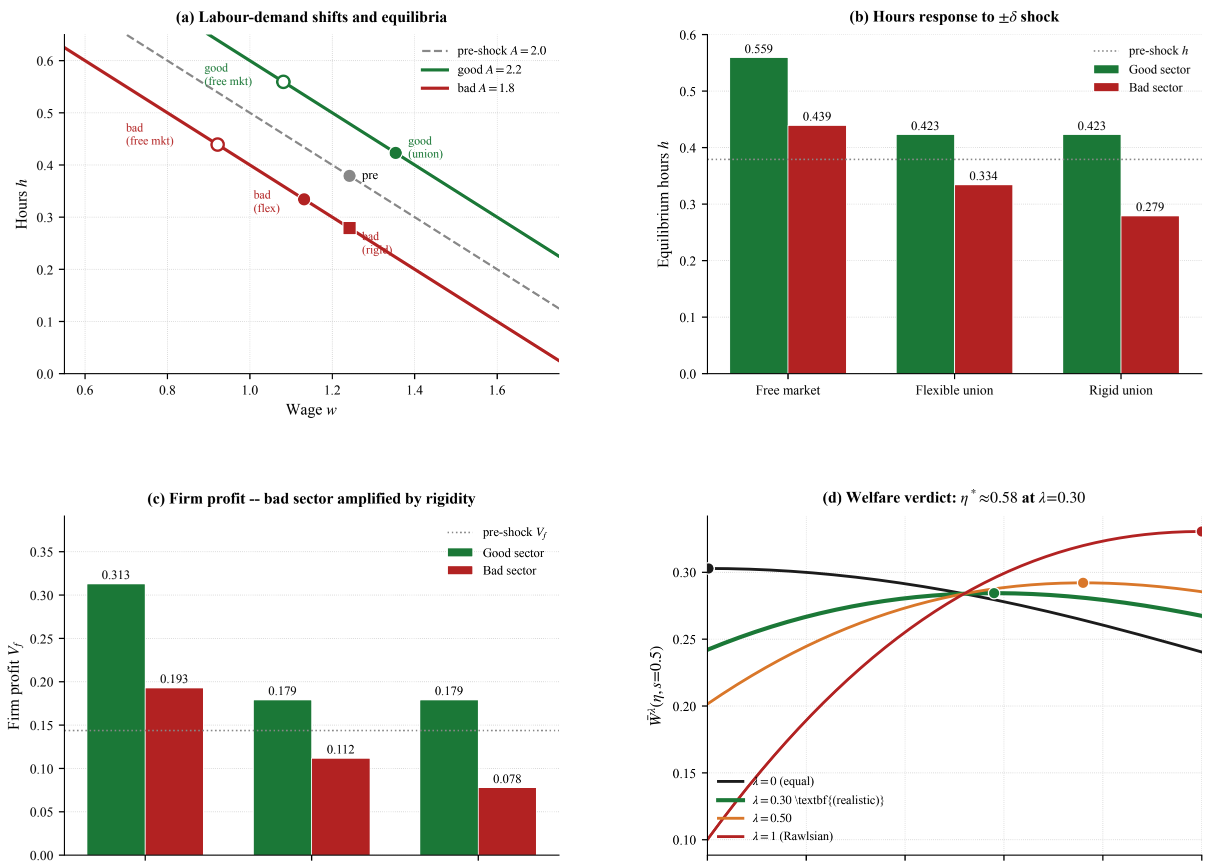

Figure 4: **Asymmetric pass-through under the three regimes, plus the criterion-dependent welfare verdict.** Pre-shock symmetric $(A, w, h) = (2, 1.242, 0.379)$ at $\eta = 0.5$; shock $\delta = 0.2$. **(a)** Labour-demand shifts and equilibria: rigid bad-sector equilibrium sits on the new demand curve at the stuck pre-shock wage. **(b)** Hours response: good-sector hours rise under all regimes (the productivity gain is real); bad-sector rigid is 36% below free market. **(c)** Firm profit: good-sector profit rises under all regimes; bad-sector rigid is 60% below free market. **(d)** *Aggregate welfare verdict is criterion-dependent.* $\bar{W}^{\lambda}(\eta, s = 0.5)$ for $\lambda \in \{0, 0.30, 0.50, 1\}$, peaks marked. At $\lambda = 0$ (equal weights): monotone-decreasing, optimum at $\eta^* = 0$. At realistic $\lambda = 0.30$ (green): interior optimum at $\eta^* \approx 0.58$ – moderately strong union welfare-optimal. *Source:* `sim_models_v2.py:make_shocks`.

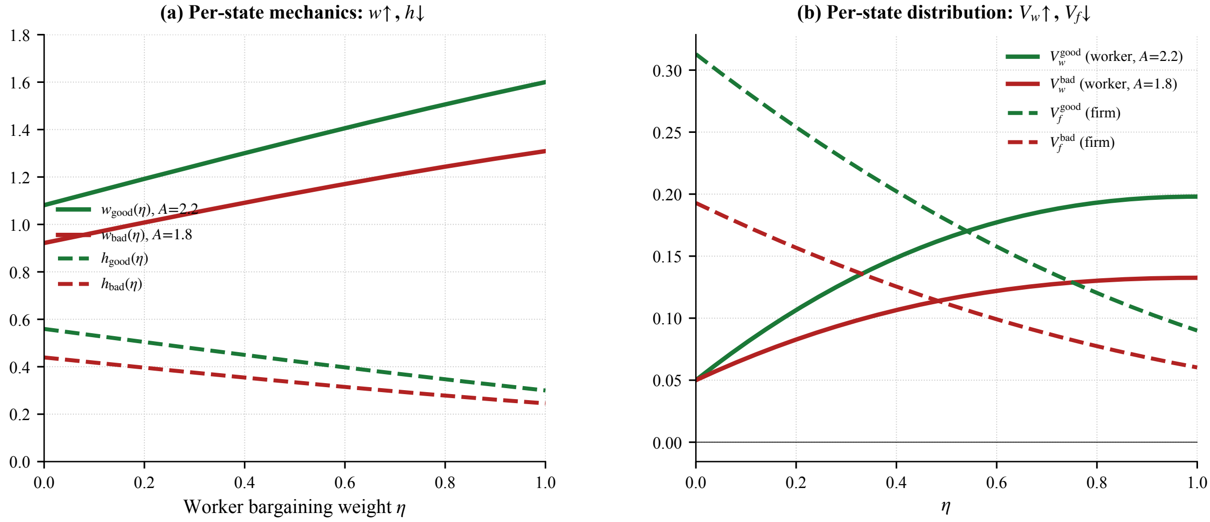

Figure 5: **Q5 RTM bargain across two sectors – underlying mechanics.** **(a)** Per-sector mechanics: $w_j(\eta)$ rises, $h_j(\eta)$ falls in each sector along its own labour-demand curve. The good sector has uniformly higher w and h. **(b)** Per-sector distribution: $V_w^j(\eta)$ rises (worker captures rents – solid lines), $V_f^j(\eta)$ falls (firm loses rents – dashed lines). Both effects are present in both sectors; magnitudes scale with A_j. *Source:* `sim_models_v2.py:make_rtm`.

5.5 Aggregate verdict: dependence on (λ, s)

The economy-wide welfare verdict turns on two dimensions: the welfare criterion λ and the sectoral composition s. Define the aggregates by linear weighting:

$$\bar{h}(s) = s\, h_{\mathrm{G}} + (1-s)\, h_{\mathrm{B}}, \qquad \bar{W}^{\lambda}(s) = s\, W_{\mathrm{G}}^{\lambda} + (1-s)\, W_{\mathrm{B}}^{\lambda}.$$

Composition channel (s). The rigidity cost $\bar{W}^{\mathrm{flex}} - \bar{W}^{\mathrm{rigid}}$ depends only on the bad sector (where rigidity binds): 0.019 at $s=0$ (all bad), 0.000 at $s=1$ (all good); linear in $1-s$. The unionisation cost $\bar{W}^{\mathrm{FM}} - \bar{W}^{\mathrm{flex}}$ is roughly constant ($\approx$0.016–0.019). **Total cost** of unionised rigidity vs. free market: 0.035 at $s=0$, falling to 0.019 at $s=1$ – declining-sector economies pay nearly twice the welfare cost of expanding-sector economies.

Criterion channel (λ). Inequality aversion attenuates the rigidity cost. In the rigid bad sector the worker's preserved wage delivers $V_w = 0.130$ (vs. flex $V_w = 0.115$), worth $1+\lambda = 1.3$ at $\lambda = 0.30$; the firm's profit collapse $V_f = 0.078$ (vs. flex 0.112) is worth $1-\lambda = 0.7$. The all-bad-sector $(s=0)$ rigidity cost falls from $+0.019$ (equal weights) to $+0.005$ at $\lambda = 0.30$ – *nearly welfare-neutral* once redistribution is given welfare value, because the union member is materially better off at the stuck wage.

Headline. At realistic $\lambda = 0.30$ and any $s \in [0,1]$, the welfare-maximising bargaining strength is $\eta^* \approx 0.58$ in each sector and in the aggregate. The total welfare cost of unionised rigidity vs. free market is ~ 0.005 to ~ 0.019 across s – an order of magnitude smaller than the equal-weight estimate (0.035 at $s=0$). The union is welfare-improving on net under any plausible inequality aversion, with the residual rigidity cost concentrated in declining-sector economies.

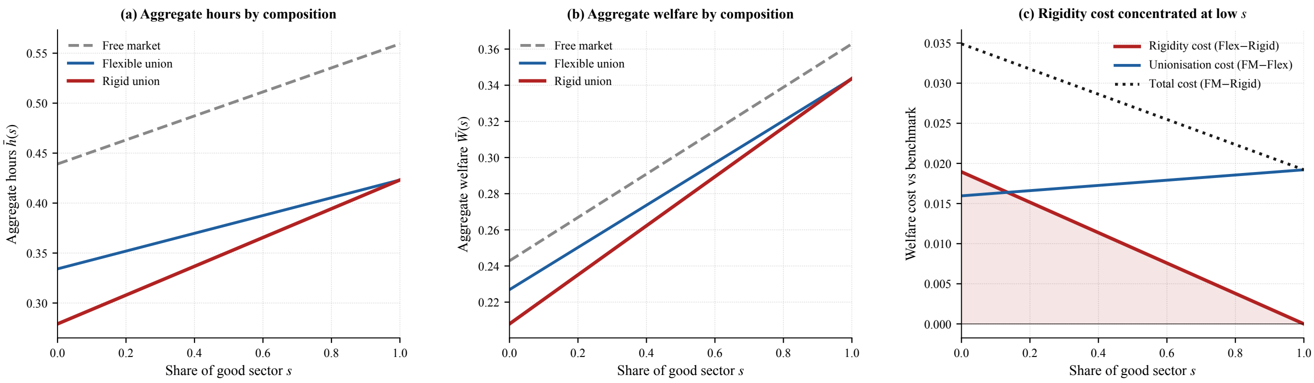

Figure 6: **Aggregate hours and welfare by sector composition s (equal-weight $\lambda = 0$).** **(a)** $\bar{h}(s)$ for the three regimes. **(b)** $\bar{W}(s)$: rigid uniformly below flexible; gap shrinks as $s \to 1$. **(c)** Welfare-cost decomposition: rigidity cost concentrates at low s (linear in $1-s$); unionisation cost roughly constant; total cost falls from ~ 0.035 at $s=0$ to ~ 0.019 at $s=1$. Under $\lambda = 0.30$ the rigidity-cost curve shrinks to ~ 0.005 at $s=0$. *Source:* `sim_models_v2.py:make_aggregation`.

6. Q7 – Welfare comparison: institutional channels

Question 7. *Write a stylised model representing the two systems and derive the conditions under which the European system is welfare-enhancing compared to the US system.*

The labour-market welfare comparison rests on two institutional channels already characterised: the union bargain of §5 and the hours regulation of §4. The US system is the firm-power baseline ($\eta = 0$, no cap); the EU system layers a positive bargaining weight $\eta^{EU} = 0.5$ and a binding regulation $\bar{h} = h^{\mathrm{soc}} = 0.40$.

6.1 Welfare condition

By the Q4 result (2), EU welfare-dominates iff its compounded hours $h(\eta^{EU}; \bar{h}^{EU})$ lie in the band $\left(2h^{\text{soc}} - h^{\text{firm}},\, h^{\text{firm}}\right) = (0.30, 0.50)$. By the §5.5 redistribution analysis, the equal-weight verdict ($\lambda = 0$) is corner-degenerate ($\eta^* = 0$, EU loses on union); the realistic Saez–Stantcheva verdict ($\lambda = 0.30$) gives $\eta^* \approx 0.58$ and the EU advantage at full configuration is $\Delta W^\lambda = +0.154$.

6.2 Hours as a fiscal–investment input

Each unit of hours generates social returns that the static private-surplus comparison has not yet credited: tax revenue $T = \tau w h$ funds public goods G (Barro 1990; Aschauer 1989; Glomm–Ravikumar 1992); firm profit funds capital (Trabandt–Uhlig 2011); cumulative hours generate on-the-job HC (Bils–Klenow 2000) and R&D / TFP (Rachel 2020; Aghion–Howitt). A single literature-anchored elasticity $\eta_\phi \approx 0.40$ bundles these via a Barro reduced form $A^{\text{soc}}(h) = A_0[1 + \eta_\phi(h/h^{US} - 1)]$, with sub-channel contributions 0.20 public goods, 0.10 capital, 0.05 HC, 0.05 R&D. EU institutions that pull h below h^{US} incur a productivity drag $A_0\eta_\phi(h^{EU}/h^{US} - 1)h^{EU} \approx -0.046$ that compounds with the worker-capture cost.

6.3 Verdict

Q7 verdict. *(i) Equal weights ($\lambda = 0$):* EU welfare-dominated – union and cap pull h below h^{soc}, the social-productivity correction adds a ~ 0.05 penalty, US wins by ~ 0.09. *(ii) Realistic Saez–Stantcheva ($\lambda = 0.30$):* EU welfare-*dominant* by $\Delta W^\lambda \approx +0.10$ (the $+0.154$ headline of §5.5 net of the -0.05 social-productivity drag) – union-driven redistribution overwhelms the misallocation cost. *(iii) Identification caveat:* Olovsson (2009) reproduces the EU–US hours gap from welfare-state policy alone; the labour-market block is sign-determinate but quantitatively second-order relative to non-labour channels (life expectancy, consumption, inequality – Jones–Klenow 2016). The verdict's sign therefore turns on the planner's λ.

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Replication scripts in `research/`: `build_figure_panel.py` (Q1 figure); `sim_models_v2.py` (Q2 decomposition, Q5 RTM, §5.5b Saez–Stantcheva weights, §5.6 sectoral shocks, Q7 cultural expansion + layered framework, §7.5 sensitivity); `sim_mechanisms.py` (§4 cap simulation).